FRONTEND DEVELOPMENT / SE-2501

KZ Travel Assignment 4

Introduction to JavaScript

Team: KZ Travel Group: SE-2501
Prepared: 11 October 2026

Team member	JavaScript responsibility
Ramazan Utegen	Task 1: form validation, password length and confirmation
Ziyadinkhan Kudaibergenuly	Task 2: FAQ accordion
Ruzimuhammad	Tasks 3-5: contact popup, background colours and live clock

Objective

Extend the existing six-page Kazakhstan travel guide with interactive JavaScript features: input validation, expandable FAQ answers, a contact popup, background colour changes, and a readable current date and time. Keep the previous responsive Bootstrap layouts and destination content.

Shared website

https://ramazanutegen.github.io/assignment1web/

Repository: github.com/ramazanutegen/assignment1web

Pages: Home, Aktau, Astana, Almaty, Shymkent, and About Us. Every footer names all three members. JavaScript is linked with `defer`, and shared files run after the page markup is parsed.

Team preparation

The table sets responsibility and study ownership for the shared implementation. Each member must personally review and practise their component, understand their teammates' code, submit the report and ZIP, and defend the project during the practice lesson.

Task 1 / Form validation

Responsible member: Ramazan Utegen
Location: Home / Community signup; shared validation on existing enquiry forms.

1. Add required name, email, message, password, and confirmation inputs. Choose and display a minimum password length of eight characters.

2. Select forms with `data-demo-form` and attach a submit listener in `js/forms.js`. Prevent normal submission and check the entries in JavaScript.

3. Reject empty or whitespace-only required values, invalid email syntax, short passwords, and mismatched confirmation. Require the contact form's acknowledgement checkbox.

4. Show an inline error beside each invalid field and a summary. Set accessible error attributes and focus the first invalid input.

5. Recheck corrected fields, including confirmation when the original password changes. Show success for valid input and clear passwords after a successful check.

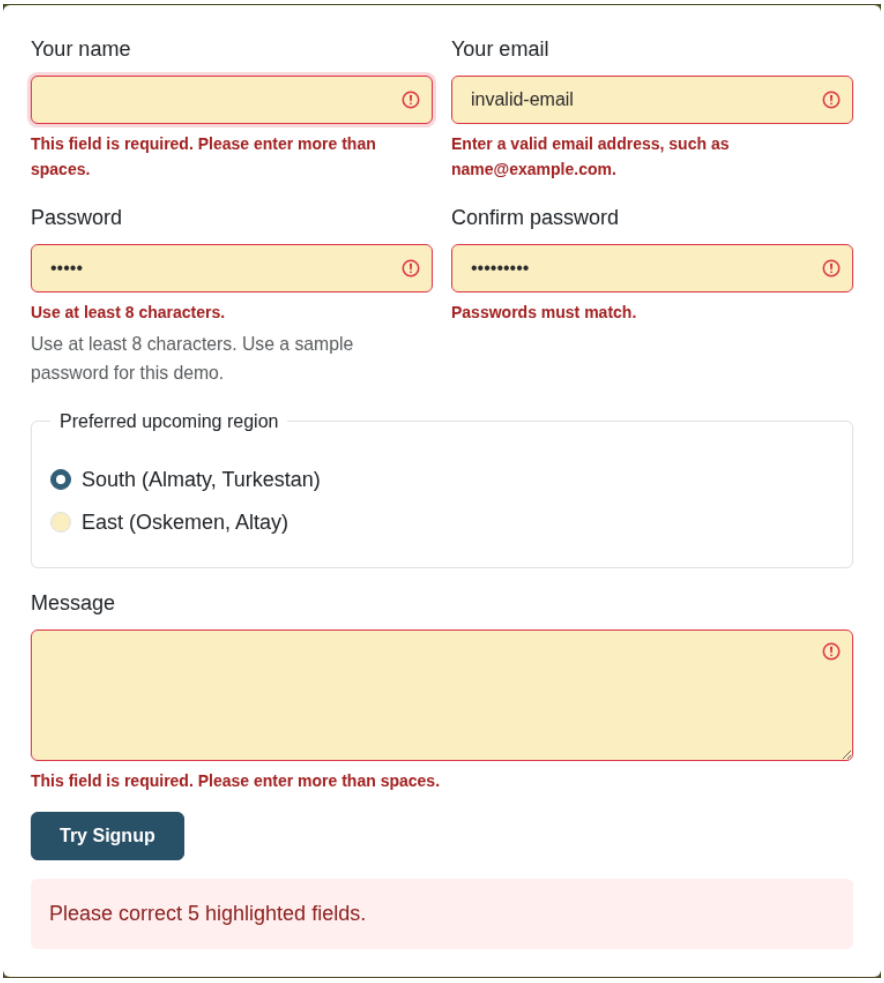


Figure 1. Required-field, email, length, and matching-password errors.

Forms validate in the browser only. They do not transmit data or create accounts, bookings, messages, or subscriptions.

Task 2 / FAQ accordion

Responsible member: Ziyadinkhan Kudaibergenuly
Location: Home / Travel FAQ.

1. Create four question buttons and matching answer panels with unique IDs. All answers start closed.

2. In `js/accordion.js`, toggle `aria-expanded`, answer visibility, `inert`, and the item's open class on click.

3. Animate the CSS grid row from 0fr to 1fr over 0.3 seconds. Disable transitions when reduced motion is requested.

4. Verify repeated clicks and keyboard Enter open and close each question independently.

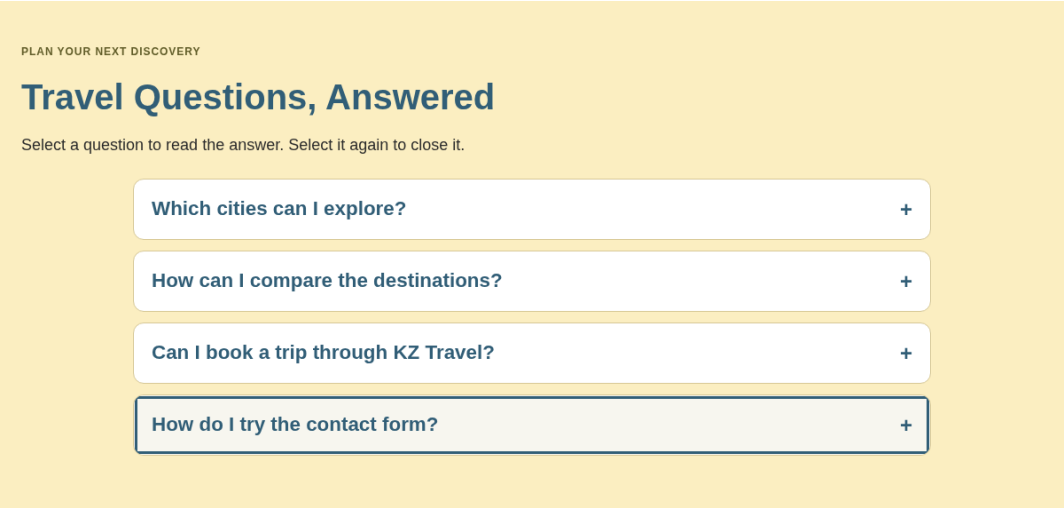


Figure 2. Default state: all four answers are hidden.

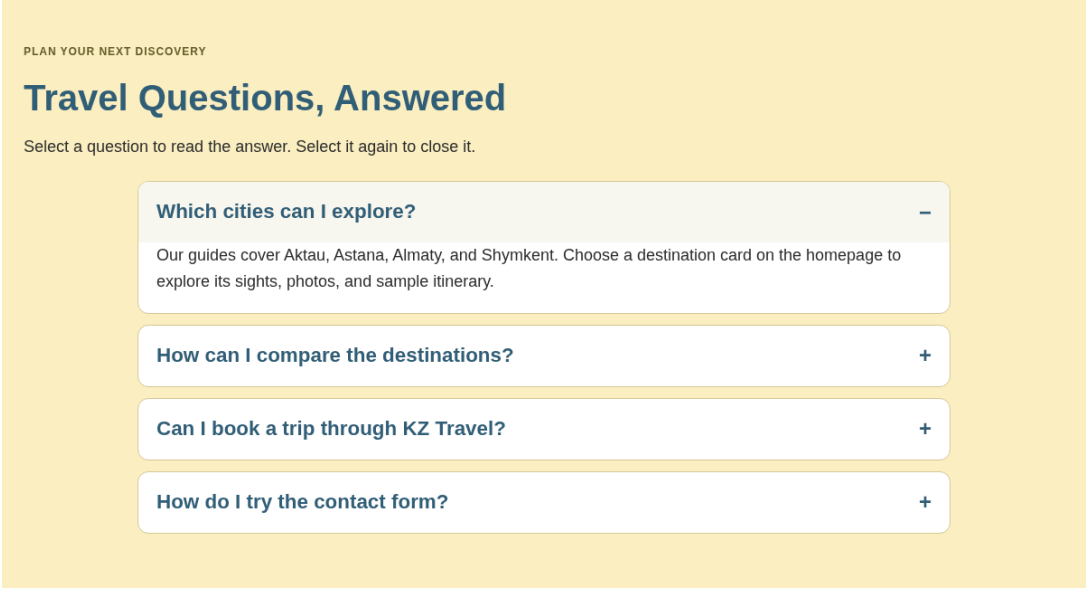


Figure 3. Clicking the first question reveals its answer.

Task 3 / Popup contact form

Responsible member: Ruzimuhammad
Location: About Us / Contact the Team.

1. Place the contact form inside an initially closed, labelled HTML dialog.

2. Call `showModal()` from Contact Us and show a dark backdrop. Keep its width and height within the viewport.

3. Close it with the close button, Escape, or an outside click. Compare pointer-down and click positions to avoid accidental dismissal while dragging from inside.

4. Keep keyboard focus inside while open, stop background scrolling, and restore focus to Contact Us after closing. Apply shared form validation.

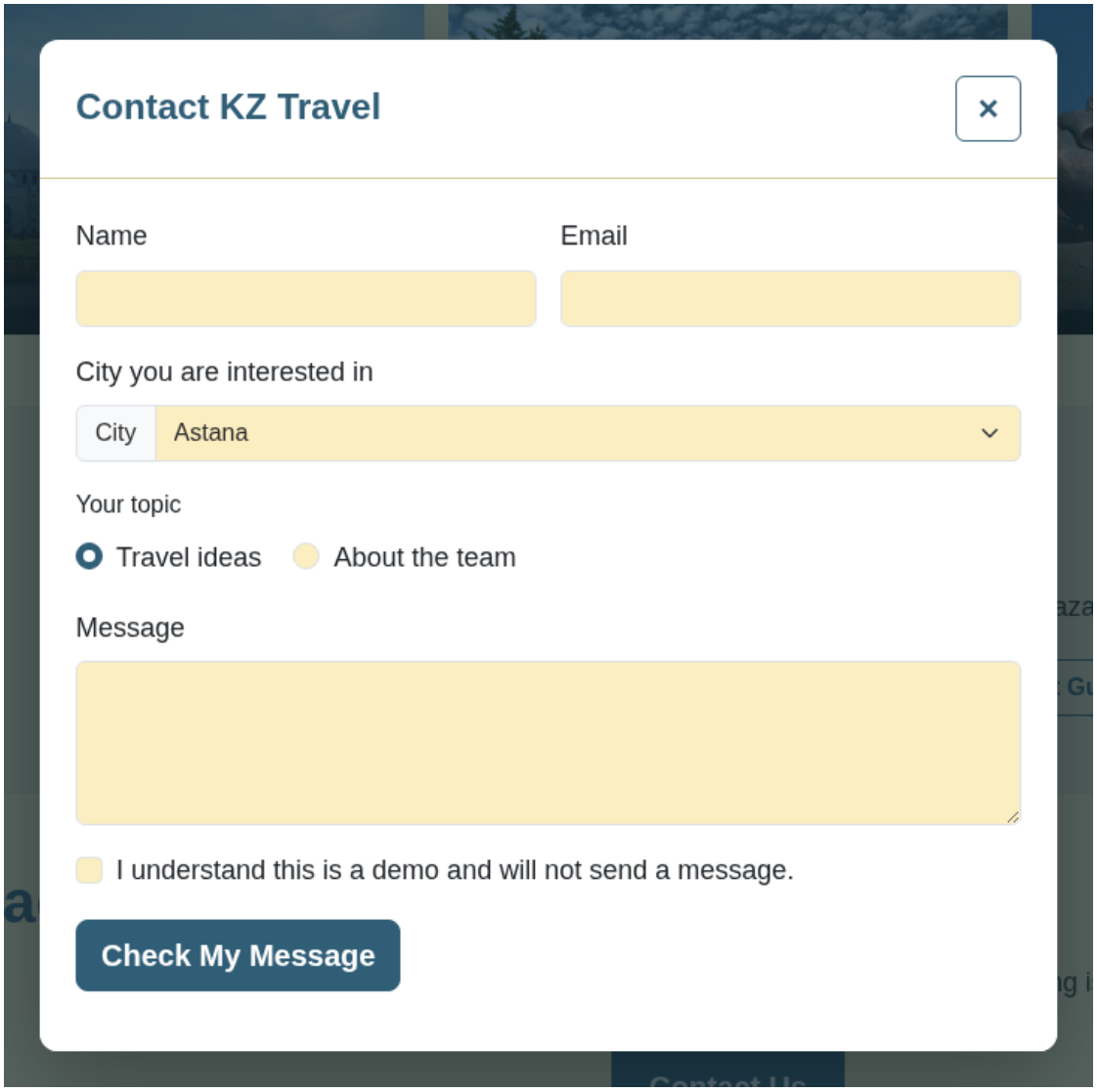


Figure 4. The contact form opens above the page with a close control and backdrop.

The popup was checked at desktop and phone sizes in Chromium and Firefox. All three close methods and valid contact input passed.

Task 4 / Change background colour

Responsible member: Ruzimuhammad Location: Toolbar on all six pages.

1. Add a Change background button and a current-colour label.

2. Store Sand, Sky, and Steppe in an array in `js/preferences.js`. Advance with `(index + 1) % colours.length`.

3. Assign the selected colour to `document.body.style.backgroundColor` and update the label. Use a short CSS transition and a reduced-motion override.



Figure 5. Original Sand page background.



Figure 6. The button changes the background and label to Sky.

Task 5 / Display current date and time

Responsible member: Ruzimuhammad Location: Toolbar on all six pages.

1. Create a Date object and update the time element's machine-readable datetime attribute.

2. Use Intl.DateTimeFormat for a readable English date and time with seconds. Use Asia/Almaty and label UTC+5 explicitly.

3. Update immediately, then every 1,000 milliseconds with setInterval in `js/clock.js`.

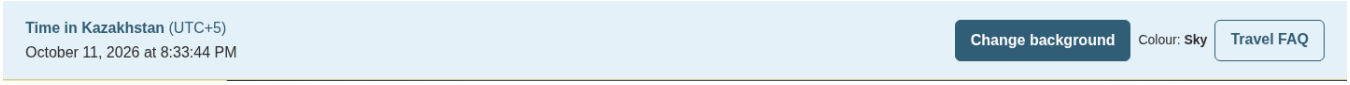


Figure 7. Readable Kazakhstan time, updated every second.

The clock was tested with the browser configured to America/New_York: it still displayed Kazakhstan time, and its datetime value advanced automatically.

Deployment and verification

1. Run the website locally with `python3 -m http.server 8000`. Check assets, unique IDs, and internal fragment links.

2. Exercise all six pages and all five features using the browser integration check in `scripts/verify-assignment4.cjs`.

3. Commit and push to the existing shared repository on main. GitHub Pages builds the root of main; verify the build and the updated files at the live website URL.

4. Generate this PDF, inspect its rendered pages, and package the website, assets, report, screenshots, and study notes into the project ZIP.

Browser	Viewport	Result
Chromium	1440 x 1100 / desktop	Passed
Chromium	390 x 844 / phone size	Passed
Firefox	1440 x 1100 / desktop	Passed
Firefox	390 x 844 / phone size	Passed

Checks: missing and whitespace-only fields; email syntax; short and mismatched passwords; successful submission; FAQ clicks and keyboard toggling; popup opening, closing, and focus; complete colour cycling; live clock updates and timezone; footer names; overflow; and JavaScript/HTTP asset errors.

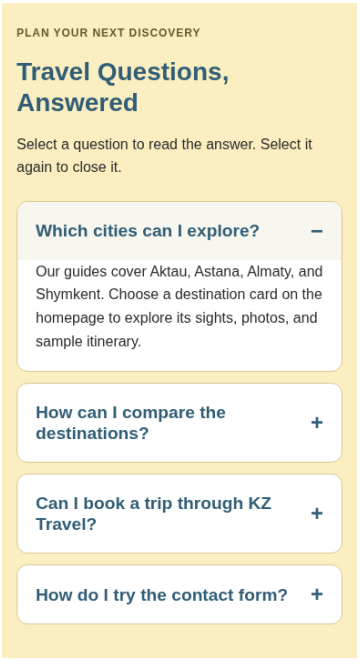


Figure 8. Phone-size FAQ.

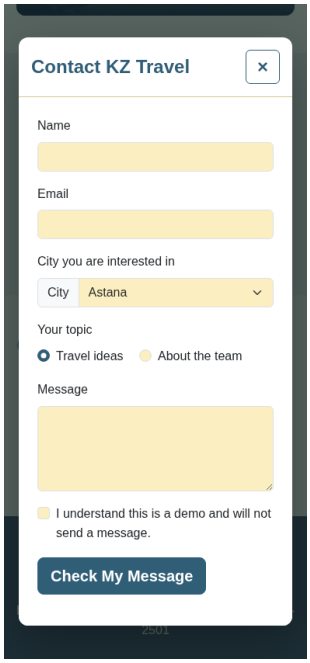


Figure 9. Phone-size popup.

Reflection and defence preparation

JavaScript adds immediate input feedback, expandable content, a popup contact form, background control, and a live clock to the existing travel guide. Shared scripts make the behaviour consistent across pages while retaining the previous responsive design. The tasks connect DOM selection, events, conditionals, arrays, CSS changes, and timers to visible interactions.

Each member's practice task

Member	Practice change and explanation
Ramazan Utegen	Change the minimum password length and hint. Explain required-field checks, email syntax, confirmation, and preventDefault.
Ziyadinkhan Kudaibergenuly	Add a fifth FAQ with unique IDs. Explain DOM selection, aria-expanded, open classes, and the CSS transition.
Ruzimuhammad	Add a fourth background colour and change the clock format. Explain modal close methods, focus, arrays, Date, and setInterval.

All members should also explain another member's component and demonstrate the full website. Study `docs/assignment-4.md` for the complete walkthrough and additional screenshots.

Final submission checklist

• All five requested JavaScript components are included, linked, and tested.

• All three team members are named in every HTML footer.

• The report includes team/group details, an objective, task steps, screenshots, the deployed URL, verification, and reflection.

• Every member submits the same report and project ZIP individually in the LMS.

• Check the Assignment 4 deadline in the LMS; no specific date is provided in the assignment PDF.

• Attend and defend the work during the practice lesson. Defence is worth 40%; implementation is 50% and the report is 10%.

The forms remain static demonstrations. A production service would need server-side processing before creating accounts or sending messages. The next learning step is for each member to rehearse the implementation and practise making small changes independently.